

Engineering Chemistry Lab

Course Code: 2079

Semester: 2

Credits: No Credit

Dept: Chemical / Food Tech

[Official Syllabus PDF](#)

Official Source Declaration

This lesson is architected based on the official **SITTTR Kerala REV2021** syllabus for Course Code **2079 (Engineering Chemistry Lab)**. All modules, radicals, and assessment criteria are derived directly from the authoritative document provided for Diploma in Chemical Engineering and Food Processing Technology.

Note: While the syllabus lists "No Credit", the course involves 45 contact hours and two internal Lab Examinations (Lab Exam I & II) which are critical for semester progression.

Course Overview

Teaching Scheme

- **L:T:P** — 3:0:0 (Practical oriented)
- **Total Periods** — 45 per semester
- **Category** — Engineering Science

Course Objectives

To provide hands-on experience in the qualitative analysis of Inorganic compounds as simple salts and binary mixtures.

Malayalam support:

ഈ കോഴ്സ് വിദ്യാർത്ഥികൾക്ക് അനോറിക് സംയുക്തങ്ങൾ (Inorganic compounds) ക്വണ്ടിറ്റേറ്റീവ് അനലിസിസ് (Qualitative Analysis) നടത്താൻ സഹായിക്കും. വിദ്യാർത്ഥികൾക്ക് ലളിതമായ ലവണങ്ങൾ (Simple Salts) ക്വണ്ടിറ്റേറ്റീവ് അനലിസിസ് നടത്താൻ സഹായിക്കും. വിദ്യാർത്ഥികൾക്ക് ബൈനറി മിശ്രിതങ്ങൾ (Binary Mixtures) ക്വണ്ടിറ്റേറ്റീവ് അനലിസിസ് നടത്താൻ സഹായിക്കും.

Course Outcomes (CO)

CO No.	Description	Duration	Cognitive Level
CO1	Analyse acid radicals	9 Hours	Applying
CO2	Analyse basic radicals	10 Hours	Applying
CO3	Analyse simple salts	12 Hours	Applying

CO No.	Description	Duration	Cognitive Level
CO4	Analyse binary mixtures	14 Hours	Applying

Module 1: Analysis of Acid Radicals (Anions)

9 Hours

Acid radicals are the negatively charged ions in an inorganic salt. The analysis involves systematic identification using preliminary and confirmatory tests.

Scope of Radicals

The syllabus requires identification of the following acid radicals:

- Carbonate (CO_3^{2-})
- Sulphide (S^{2-})
- Sulphite (SO_3^{2-})
- Sulphate (SO_4^{2-})
- Thiosulphate ($\text{S}_2\text{O}_3^{2-}$)
- Nitrate (NO_3^-)
- Fluoride (F^-)
- Chloride (Cl^-)
- Bromide (Br^-)
- Iodide (I^-)
- Acetate (CH_3COO^-)
- Oxalate ($\text{C}_2\text{O}_4^{2-}$)
- Phosphate (PO_4^{3-})
- Borate (BO_3^{3-})
- Chromate (CrO_4^{2-})

Systematic Procedure for Acid Radicals

1. **Action of Dilute HCl:** Carbonate (brisk effervescence), Sulphide (rotten egg smell), Sulphite (burning sulphur smell).
2. **Action of Conc. H_2SO_4 :** Chloride (pungent fumes), Nitrate (brown fumes with copper turnings), Acetate (vinegar smell), Oxalate ($\text{CO} + \text{CO}_2$ evolution).
3. **Special Tests:** Sulphate (BaCl_2 test), Phosphate (Ammonium molybdate test), Borate (Ethyl borate test).

Malayalam support:

ഓക്സൈഡ് അയോണുകൾക്ക് ഓക്സൈഡേഷൻ സംഭവിക്കുന്നു. HCl (Dilute HCl) ഉപയോഗിച്ച് പരീക്ഷിക്കുക. ഓക്സൈഡ് അയോണുകൾക്ക് ഓക്സൈഡേഷൻ സംഭവിക്കുന്നു. H_2SO_4 (Conc. H_2SO_4) ഉപയോഗിച്ച് പരീക്ഷിക്കുക. ഓക്സൈഡ്, ഓക്സൈഡേഷൻ സംഭവിക്കുന്നു. ഓക്സൈഡ് അയോണുകൾക്ക് ഓക്സൈഡേഷൻ സംഭവിക്കുന്നു.

Module 2: Analysis of Basic Radicals (Cations)

10 Hours

Basic radicals are positively charged ions. They are classified into groups based on their solubility products and identified using group reagents.

Group Classification

Group	Radicals	Group Reagent
Group 0	Ammonium (NH_4^+)	NaOH
Group I	Silver (Ag^+), Lead (Pb^{2+}), Mercury (Hg_2^{2+})	Dilute HCl
Group II	Copper (Cu^{2+}), Cadmium (Cd^{2+}), Bismuth (Bi^{3+}), Arsenic (As^{3+}), Antimony (Sb^{3+}), Tin ($\text{Sn}^{2+}/\text{Sn}^{4+}$)	H_2S gas in presence of Dil. HCl
Group III	Iron (Fe^{3+}), Aluminium (Al^{3+}), Chromium (Cr^{3+})	NH_4OH in presence of NH_4Cl
Group IV	Zinc (Zn^{2+}), Manganese (Mn^{2+}), Cobalt (Co^{2+}), Nickel (Ni^{2+})	H_2S gas in presence of NH_4OH
Group V	Barium (Ba^{2+}), Strontium (Sr^{2+}), Calcium (Ca^{2+})	$(\text{NH}_4)_2\text{CO}_3$ in presence of NH_4Cl & NH_4OH
Group VI	Magnesium (Mg^{2+}), Sodium (Na^+), Potassium (K^+)	Specific reagents (e.g., Na_2HPO_4 for Mg)

Module 3: Analysis of Simple Salts

12 Hours

A simple salt consists of one acid radical and one basic radical. The analysis must follow a systematic workflow:

1. Preliminary Examination (Color, State, Solubility).
2. Identification of Acid Radical.
3. Confirmatory test for Acid Radical.
4. Preparation of Original Solution (OS).
5. Identification of Basic Radical Group.
6. Identification of Basic Radical within the group.

7. Confirmatory test for Basic Radical.

Syllabus Constraint: The simple salt should be soluble in water, dilute acids, or conc. HCl and should not involve interfering radicals (like Fluoride, Borate, Oxalate, Phosphate) that complicate group separation.

Module 4: Analysis of Binary Mixtures

14 Hours

Binary mixtures contain two simple salts, meaning two acid radicals and two basic radicals must be identified.

Strategy: In binary mixtures, one must check if the presence of one radical interferes with the test for another. For example, in a mixture of Chloride and Bromide, special separation is needed.

Open Ended Experiments (3 Hours)

Students are encouraged to apply their analytical skills to unknown samples or real-world materials like water hardness or soil minerals (qualitative aspect).

Formula Bank & Calculations

Solubility Product (K_{sp})

The principle behind group separation. A precipitate forms when the Ionic Product exceeds K_{sp} .

Concept: For a salt A_xB_y ,
 $K_{sp} = [A]^x [B]^y$

Common Ion Effect

Used to suppress the ionization of weak electrolytes. Example: NH_4Cl suppresses NH_4OH ionization to ensure only Group III hydroxides precipitate.

Molarity Calculator (For Reagent Prep)

Mass of Solute (g)

e.g. 40

Molar Mass (g/mol)

e.g. 40 for NaOH

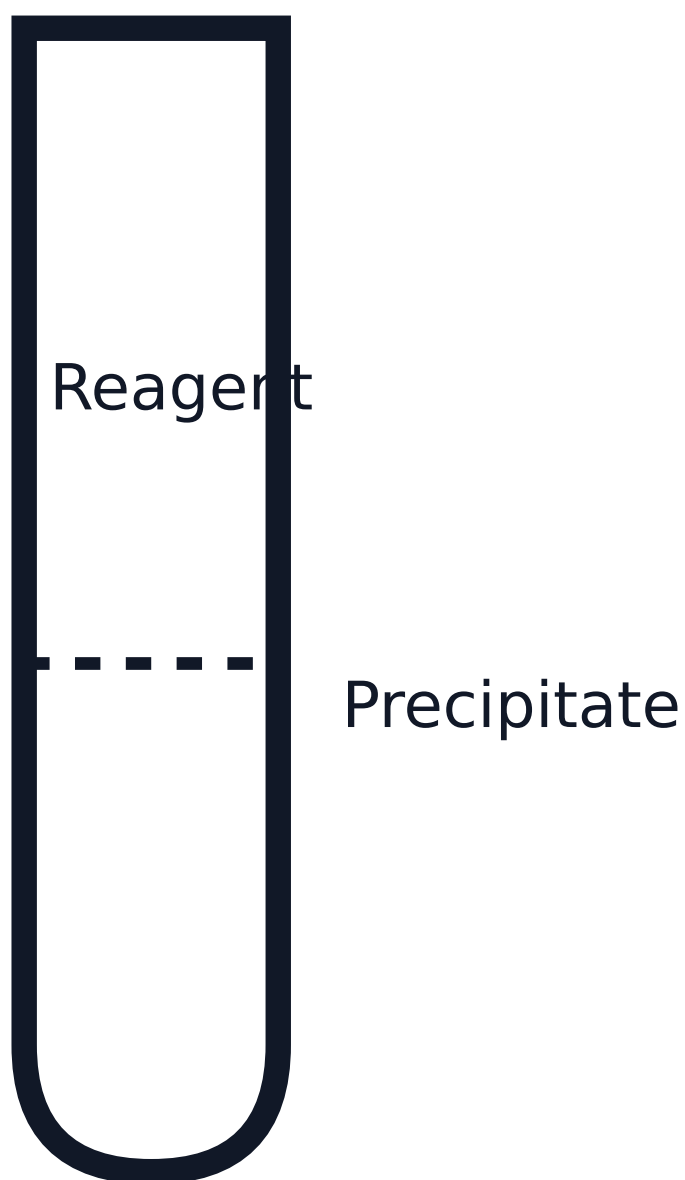
Volume of Solution (mL)

e.g. 1000

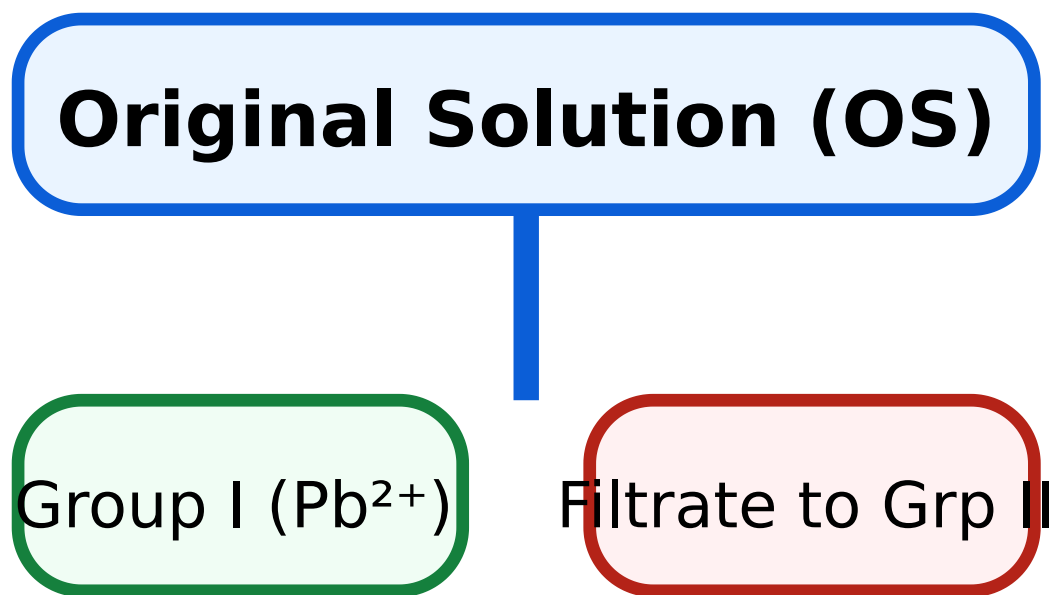
Calculate Molarity

Result: -- M

Visual Library



Basic setup for qualitative precipitation tests.



Logic of systematic group separation.

Practice Questions & Answers

1. Why is NH_4Cl added before NH_4OH in Group III analysis?

To suppress the concentration of OH^- ions through the **Common Ion Effect**. This ensures that the ionic product of Group IV, V, and VI hydroxides does not

exceed their solubility products, preventing them from precipitating along with Group III.

2. How do you identify the presence of Carbonate radical?

Treat the salt with dilute HCl. Brisk effervescence occurs due to the evolution of CO₂ gas. When this gas is passed through lime water, it turns milky due to the formation of Calcium Carbonate.

3. What is the confirmatory test for Nitrate?

Brown Ring Test: To the salt solution, add freshly prepared Ferrous Sulphate solution. Then, carefully add concentrated H₂SO₄ along the sides of the test tube. A brown ring forms at the junction of the two liquids.

Practice Model Paper

PRACTICE MODEL EXAM — NOT AN OFFICIAL PAPER

Time: 3 Hours | Max Marks: 100 (Internal Pattern)

Part A: Preliminary (20 Marks)

1. Perform physical examination and solubility test for the given salt mixture.
2. Report the color, state, and solubility in different solvents.

Part B: Acid Radical Analysis (30 Marks)

3. Identify and confirm two acid radicals present in the given binary mixture with systematic steps.

Part C: Basic Radical Analysis (40 Marks)

4. Identify and confirm two basic radicals present in the given binary mixture using group separation.

Part D: Viva & Record (10 Marks)

5. Answer questions on group reagents and solubility products.

Quick Revision

Group Reagents

- Gr I: Dil. HCl
- Gr II: $\text{H}_2\text{S} + \text{HCl}$
- Gr III: $\text{NH}_4\text{Cl} + \text{NH}_4\text{OH}$

Flame Test Colors

- Barium: Apple Green
- Strontium: Crimson Red
- Calcium: Brick Red

Common Mistakes

- Not using freshly prepared FeSO_4 for Nitrate test.
- Adding too much NH_4OH in Group III.

Glossary

- **Qualitative Analysis:** Identification of the constituents of a substance.
- **Precipitate:** An insoluble solid that emerges from a liquid solution.
- **Original Solution (OS):** A clear solution of the salt in a suitable solvent (Water/HCl).
- **Radical:** An atom or group of atoms carrying a charge.

References

- Practical Chemistry for B.Sc Main - A.O. Thomas and Mani.
- NCERT Laboratory Manual - Chemistry (Class XI & XII).
- SITTTTR Kerala Official Syllabus Revision 2021.

മലയാളം പിന്തുണ (Malayalam Support)

ഈ പരീക്ഷയിൽ ഉൾപ്പെട്ട പ്രധാനപ്പെട്ട പദങ്ങൾക്ക് മലയാളം പിന്തുണ:

- **Original Solution (OS):** മൂലദ്രവ്യം അല്ലെങ്കിൽ ലായനി. ഇത് പരീക്ഷയിൽ ഉപയോഗിക്കേണ്ടതാണ്.
- **Group Separation:** ഗ്രൂപ്പ് വേർതിരിവ്. ഇത് പരീക്ഷയിൽ ഉപയോഗിക്കേണ്ടതാണ്.

